Homework 2: Physics-Based Simulation and Machine Learning Surrogate for Tilt Dynamics

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1 Introduction

The objective of this assignment is to develop a physics-based simulator for a dynamic system, generate synthetic data, train a machine learning (ML) surrogate model, and evaluate the surrogate's ability to approximate the original simulator.

In this work, we model vehicle tilt dynamics using a proportional-derivative (PD) control strategy. The primary objective is to minimize the lateral acceleration felt by passengers during cornering. A neural network surrogate is trained to approximate the simulator output and is further used for parameter analysis and optimization.

2 Code Repository

The complete implementation of this project, including the simulator, machine learning model, dataset, and results, is available at:

`https://github.com/kgupt0310/AV_Assignments.git`

This repository provides all necessary files to reproduce the results presented in this report.

3 Governing Equations

The lateral acceleration experienced by a vehicle in a turn is given by:

$$a_{lat} = \frac{v^2}{R} \tag{1}$$

The desired tilt angle is:

$$\phi_{des} = \tan^{-1}\left(\frac{a_{lat}}{g}\right) \tag{2}$$

The felt lateral acceleration experienced by the passenger is:

$$a_{felt} = \frac{v^2}{R} - g\sin(\phi) \tag{3}$$

The tilt dynamics are modeled using a PD control law:

$$\ddot{\phi} = k_p(\phi_{des} - \phi) - k_{rate}\dot{\phi} \tag{4}$$

where k_p represents the proportional gain and k_{rate} represents the damping gain.

4 Physics-Based Simulator

The system dynamics are simulated by numerically integrating the governing differential equation using an ODE solver. The simulator produces time histories of tilt angle and felt acceleration.

The primary output used for analysis is the maximum felt lateral acceleration:

$$a_{felt,max} = \max|a_{felt}(t)| \tag{5}$$

Simulation results show that the system converges to the desired tilt angle, ensuring minimal steady-state lateral acceleration.

5 Synthetic Data Generation

A dataset of approximately 10,000 samples was generated by randomly sampling the input space:

- Velocity $v \in [5, 30]$ m/s
- Turn radius $R \in [20, 100]$ m
- Proportional gain $k_p \in [2, 10]$
- Rate gain $k_{rate} \in [0.5, 5]$

Each sample consists of the input parameters and the corresponding maximum felt lateral acceleration obtained from the simulator.

6 Machine Learning Surrogate

A feedforward neural network (MLPRegressor) was used to learn the mapping:

$$(v, R, k_p, k_{rate}) \rightarrow a_{felt,max} \tag{6}$$

The dataset was split into training and testing sets, and input features were normalized using standard scaling. Model performance was evaluated using Mean Absolute Error (MAE) and Root Mean Square Error (RMSE), both of which indicated high prediction accuracy.

7 Comparison Between Simulator and ML Model

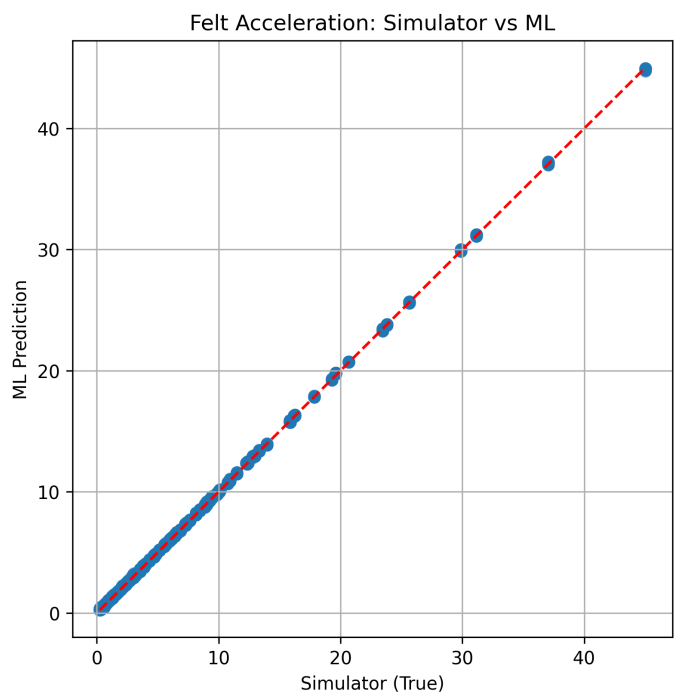

Figure 1: Scatter plot comparing ML predictions with simulator outputs. The close alignment with the diagonal indicates strong agreement.

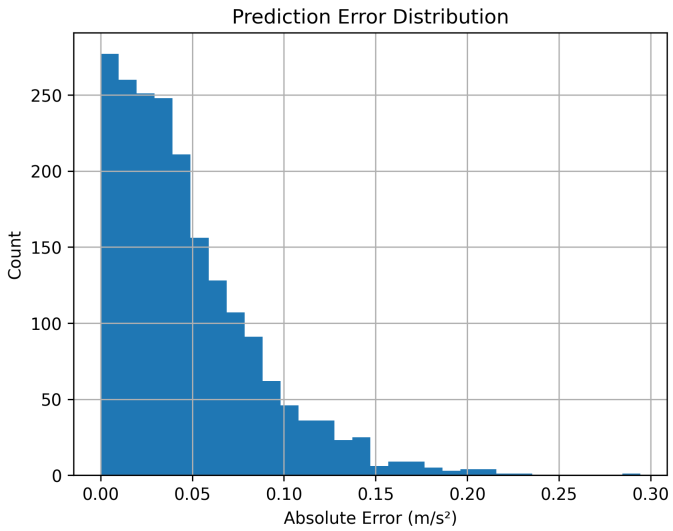

Figure 2: Distribution of prediction errors. The majority of errors are small, indicating accurate approximation.

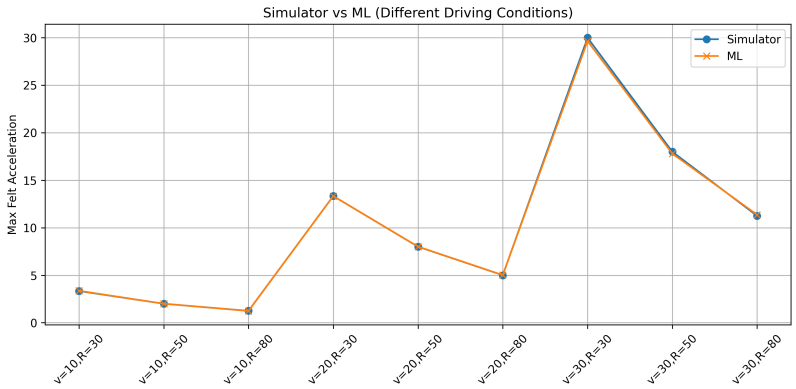

Figure 3: Comparison of simulator and ML outputs across multiple test cases. The ML model closely follows the simulator.

The ML model successfully reproduces the simulator behavior, demonstrating that it has learned the underlying input-output relationship.

8 Parameter Sensitivity Analysis

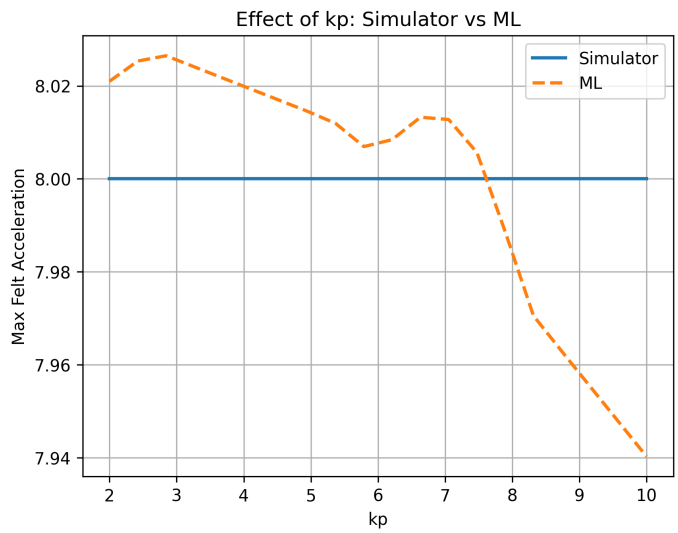

Figure 4: Effect of proportional gain k_p on system output. The simulator output remains nearly constant due to steady-state convergence.

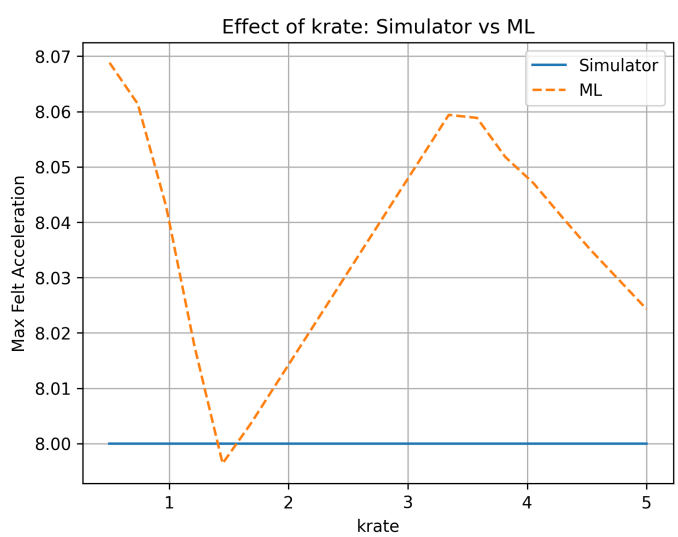

Figure 5: Effect of rate gain k_{rate} on system output. Variations are minimal, indicating limited influence on steady-state performance.

The results show that the control gains k_p and k_{rate} have minimal impact on the steady-state felt lateral acceleration. This is because the system converges to the same equilibrium regardless of gain values.

This highlights that the chosen performance metric primarily reflects steady-state behavior, while control gains mainly influence transient dynamics.

9 Optimization Using ML Surrogate

The trained ML surrogate was used to efficiently search for optimal control gains that minimize felt lateral acceleration. This approach avoids repeated numerical simulation and enables rapid evaluation of multiple parameter combinations.

10 Computational Efficiency

The computational efficiency of the simulator and ML model was evaluated by averaging execution time over multiple runs:

- Simulator average time: 5.23×10^{-4} seconds
- ML model average time: 3.72×10^{-5} seconds

The ML model is approximately 14 times faster than the simulator. This improvement arises because the simulator requires numerical integration, whereas the ML model performs direct function evaluation.

This makes the ML surrogate suitable for real-time applications such as controller tuning.

11 Discussion

The ML surrogate provides a fast and accurate approximation of the physics-based simulator. Its primary advantage is computational efficiency, especially when evaluating a large number of scenarios.

However, the ML model does not explicitly incorporate physical laws and relies entirely on the training dataset. As a result, its performance may degrade outside the sampled parameter space.

In contrast, the physics-based simulator is grounded in first principles and provides physically consistent results but at a higher computational cost.

The surrogate shows slightly higher error at extreme parameter values, where nonlinear effects are more pronounced.

The physics-based simulator has the advantage of being grounded in first principles, ensuring physically consistent results and reliable extrapolation beyond the training dataset.

The ML surrogate provides significantly faster evaluation and enables rapid parameter optimization, making it useful for real-time applications.

12 Conclusion

This work demonstrates that a machine learning surrogate can accurately approximate a physics-based simulator for vehicle tilt dynamics.

The ML model captures the nonlinear relationship between system inputs and passenger comfort while significantly reducing computation time. The results highlight the tradeoff between physical interpretability and computational efficiency, and demonstrate the usefulness of ML models in control system analysis and optimization.